

MSc Research Project/Internship/ Practicum

Research Methodology
Design and Implementation

Research Methodology

- **Scientific paper needs to be verifiable by other researchers, so that they can review the results by replicating the experiment and guaranteeing the validity.**
- **You need to give a completely accurate description of the research procedure you followed, equipment used, the technique(s), applied, set-up of scenarios/case studies run**
- **You must provide an explanation of how the raw data was compiled and analyzed used for gathering the data.**
- **Detail all the steps from data collection to final results (Process)**

Research Methodology

- It is useful for the reader to understand how you obtained your data, because it allows them to evaluate the quality of the results.
- If you were trying to obtain data about shopping preferences, you will obtain different results from a multiple-choice questionnaire than from a series of open interviews
- If the research about shopping preferences were built upon a single case study, it would have little external validity, and the reader would treat the results with the contempt that they deserve.

Research Methodology

- **The methodology can be divided into a few sections**
 - Describe the steps you followed in your research.
 - Describe the materials and equipment used in the research.
 - Explain how the samples were gathered, any randomization techniques and how the samples were prepared (e.g. cleaned and transformed)
 - Explain how the measurements were made and what calculations were performed upon the raw data.
 - Describe the statistical techniques used upon the data.

Known Issues

- Be clear on the difference between **research methodology** and **data analytics methodology**.
- CRISP-DM should be approx 1 sentence in a research methodology.

Design Specification

- **The techniques and/or architecture / framework that underlie the implementation and the associated requirements are identified and presented**
- **If a new algorithm or model is proposed, a word-based description of the algorithm/model functionality should be included**

Implementation/Solution Development

- **The purpose is to describe only the final stage of the implementation. It should describe the outputs produced, e.g. transformed data, code written, models developed, questionnaires administered.**
- **The description should also include what tools and languages you used to produce the outputs**
- **It must not contain code listing or user manual description.**